

SOFTWARE ENGINEERING LAB – 2 (UE24CS341A) Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 007

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Problem Statement 7: Hostel Maintenance & Issue Ticketing System

EPIC: Epic 1 – Ticket Logging & Issue Categorization (FR-001 and FR-005)

Note: This report covers only Epic 1, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates: Epic 2 – FR-002 & NFR-001: Sonia Sharma; Epic 3 – FR-003: Kaustubh Akash; Epic 4 – FR-004 & NFR-002: Kasissnu Ssinha.

Lab Objective

This lab converts the requirements FR-001 and FR-005 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-001 and FR-005)

The two requirements are grouped into one Epic: FR-001 defines how residents log issues with room number tags, photo evidence, and urgency categories, while FR-005 provides the Maintenance Warden with a real-time intake dashboard to oversee and prioritize active tickets.

Epic Key	Epic Name	Mapped Requirements	Description
HMT-1	Ticket Logging & Issue Categorization	FR-001 and FR-005	Enable hostel residents to submit geo-tagged maintenance tickets with photo evidence and severity tags, and provide the warden with an active intake overview.

2. User Stories for Epic 1

The Epic is broken into 6 User Stories.

Key	Story	As a...	I want...	So that...	Covers
HMT-2	1.1 Room & Block Tagging	Hostel Resident	to select my hostel block and room number from verified dropdowns	maintenance staff are directed to the exact location	FR-001

HMT-3	1.2 Issue Categorization	Hostel Resident	to select an issue category (Plumbing, Electrical, Carpentry) and initial urgency	the ticket routes to the appropriate maintenance department	FR-001
HMT-4	1.3 Photo Evidence Upload	Hostel Resident	to attach up to two photos (max 5MB each) of the broken fixture	technicians can diagnose required parts before visiting	FR-001
HMT-5	1.4 Unique Ticket Confirmation	Hostel Resident	to receive a unique Ticket ID and submission confirmation prompt	I have a verifiable tracking reference	FR-001
HMT-6	1.5 Warden Intake Dashboard	Maintenance Warden	a centralized dashboard showing all active open tickets sortable by severity and timestamp	urgent hazards are triaged first	FR-005
HMT-7	1.6 Form Auto-Drafting	Hostel Resident	partially filled form inputs to auto-save locally	accidental page refreshes do not lose my ticket data	FR-001

Acceptance criteria were taken from Lab 1. FR-001: A submitted ticket form is accepted only when block and room number tags are valid, generating a unique ID and routing to the warden queue; attachments exceeding 5MB are rejected with clear error prompts. FR-005: Warden dashboard dynamically displays all active tickets sortable by ascending/descending severity within 2 seconds.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 1 with all 6 User Stories, the priority icon and story points on each row, and the Epic panel reporting 6 work items and an estimate of 23 points.

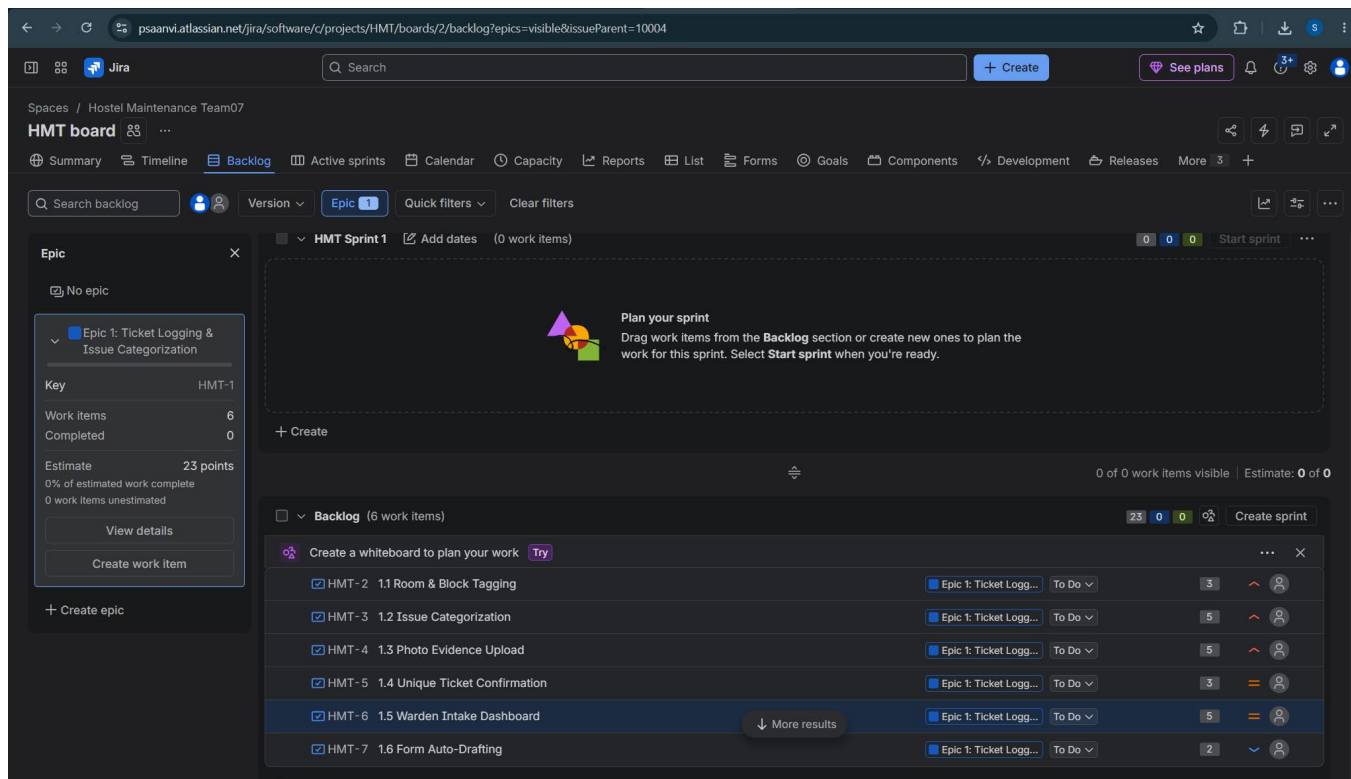


Figure 1: Jira Backlog – Epic 1 (HMT-1) with stories (3, 5, 5, 3, 5, 2 = 23 pts).

4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 1.1, 1.2, 1.3, 1.4, 1.5, 1.6.

Key	Story	Priority	Story Points
HMT-2	1.1 Room & Block Tagging	High	3
HMT-3	1.2 Issue Categorization	High	5
HMT-4	1.3 Photo Evidence Upload	High	5
HMT-5	1.4 Unique Ticket Confirmation	Medium	3
HMT-6	1.5 Warden Intake Dashboard	Medium	5
HMT-7	1.6 Form Auto-Drafting	Low	2

- 1.1, 1.2, and 1.3 (High): Core intake fields required by FR-001; without valid room locations, categories, and photos, downline dispatch cannot operate.
- 1.4 and 1.5 (Medium): Provide confirmation tracking and supervisory visibility once tickets exist.

- 1.6 (Low): Client-side persistence convenience that is not on the critical path.

Total Story Points (Epic 1): 23 | High: 13 | Medium: 8 | Low: 2

5. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

Story ID	Story Title	Saanvi	Sonia	Kaustubh	Kasissnu	Agreed
HMT-2 / 1.1	Room & Block Tagging	3	3	3	3	3
HMT-3 / 1.2	Issue Categorization	5	5	3	5	5
HMT-4 / 1.3	Photo Evidence Upload	5	5	5	8	5
HMT-5 / 1.4	Unique Ticket Confirmation	3	3	3	2	3
HMT-6 / 1.5	Warden Intake Dashboard	5	5	5	5	5
HMT-7 / 1.6	Form Auto-Drafting	2	2	3	2	2

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week)

Sprint Goal: Deliver functional ticket logging with verified room tagging, categories, and photo uploads.

Key	Story	Priority	Pts	Status
HMT-2	1.1 Room & Block Tagging	High	3	Done
HMT-3	1.2 Issue Categorization	High	5	Done
HMT-4	1.3 Photo Evidence Upload	High	5	Done

Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over

Sprint 2 (Duration: 1 week)

Sprint Goal: Provide ticket confirmation, warden dashboard sorting, and client-side draft auto-saving.

Key	Story	Priority	Pts	Status
HMT-5	1.4 Unique Ticket Confirmation	Medium	3	Done
HMT-6	1.5 Warden Intake Dashboard	Medium	5	Done
HMT-7	1.6 Form Auto-Drafting	Low	2	Done

Sprint 2 Velocity: 10 pts committed | 10 pts completed | 0 pts rolled over

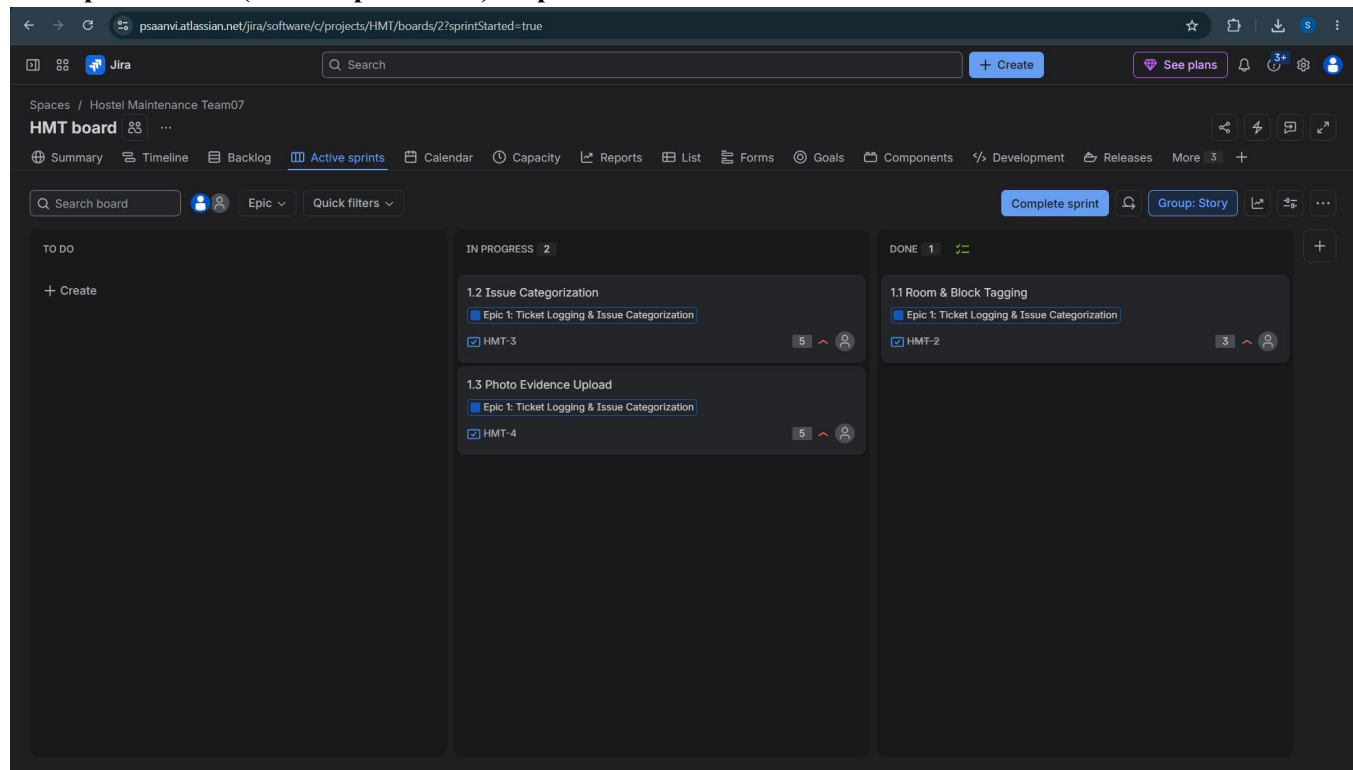
Jira Sprint Board (Active Sprint view) – Sprint 1

Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 1.1 in DONE; Stories 1.2 and 1.3 IN PROGRESS.

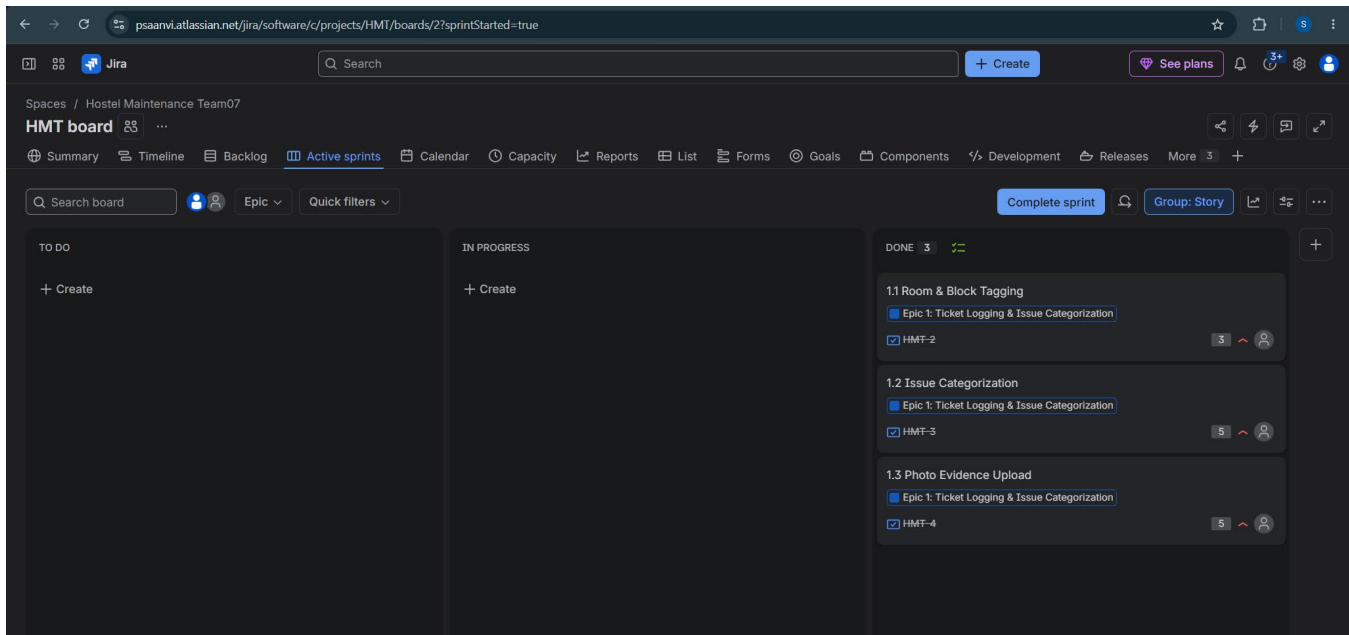


Figure 3: Sprint 1 Active Sprint board – all stories (13 pts) in DONE, before clicking Complete sprint

Sprint 2 outcome

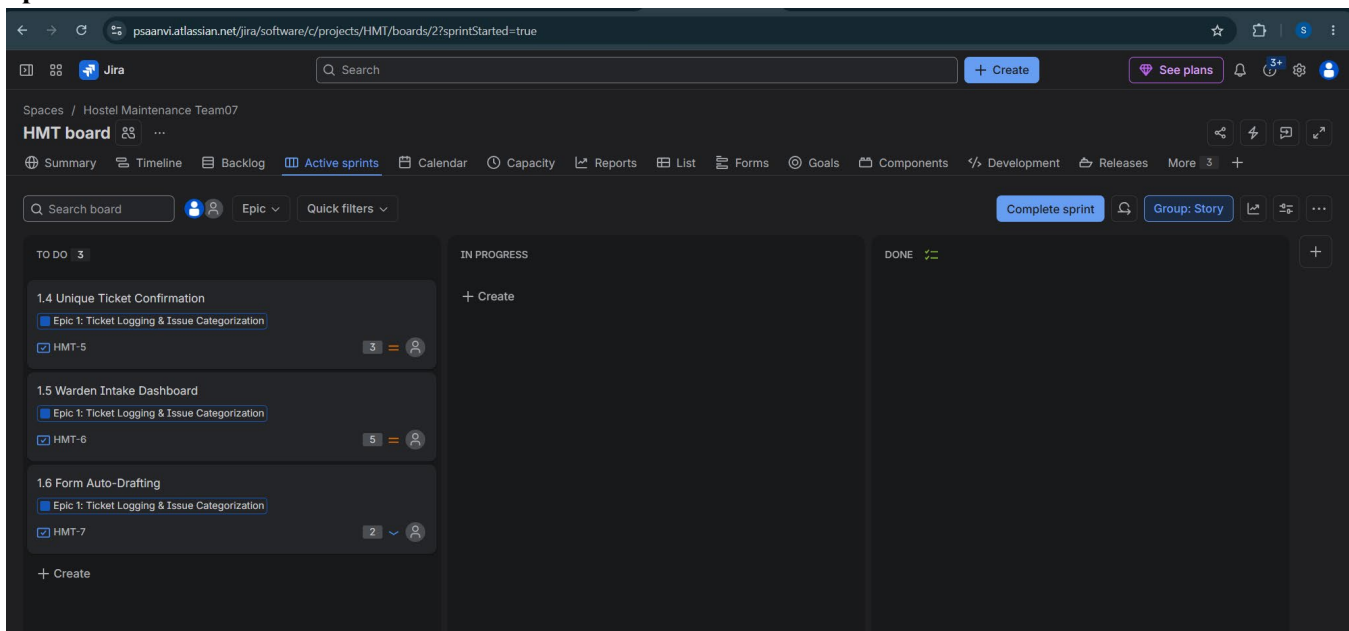


Figure 4: Sprint 2 Active Sprint board – all stories (10 pts) in TO DO.

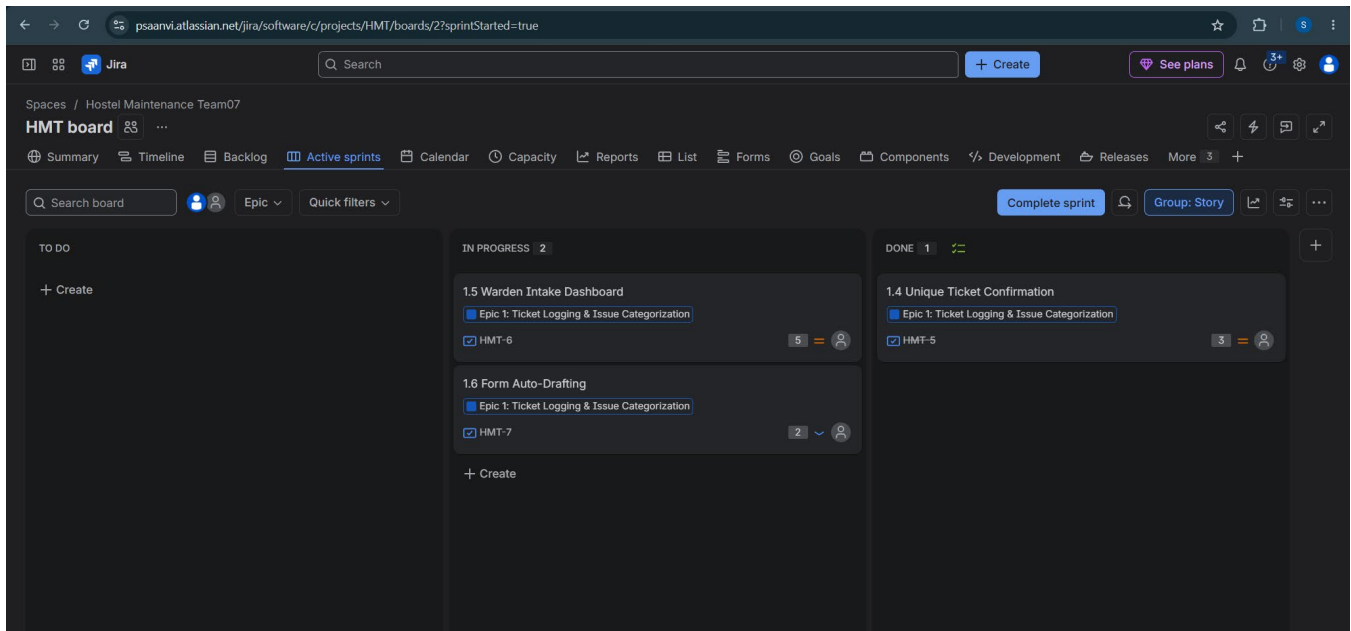


Figure 5: Sprint 2 Active Sprint board mid-sprint – Story 1.4 in DONE; Stories 1.5 and 1.6 IN PROGRESS.

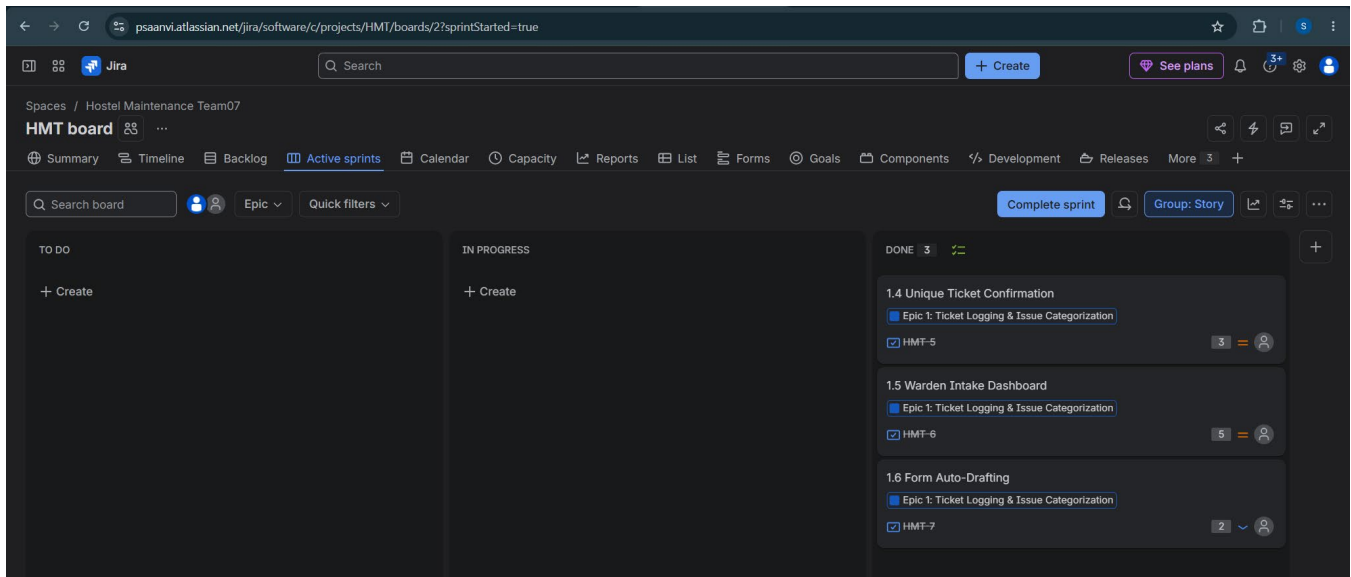


Figure 6: Sprint 2 Active Sprint board – all stories (10 pts) in DONE, before clicking Complete sprint.

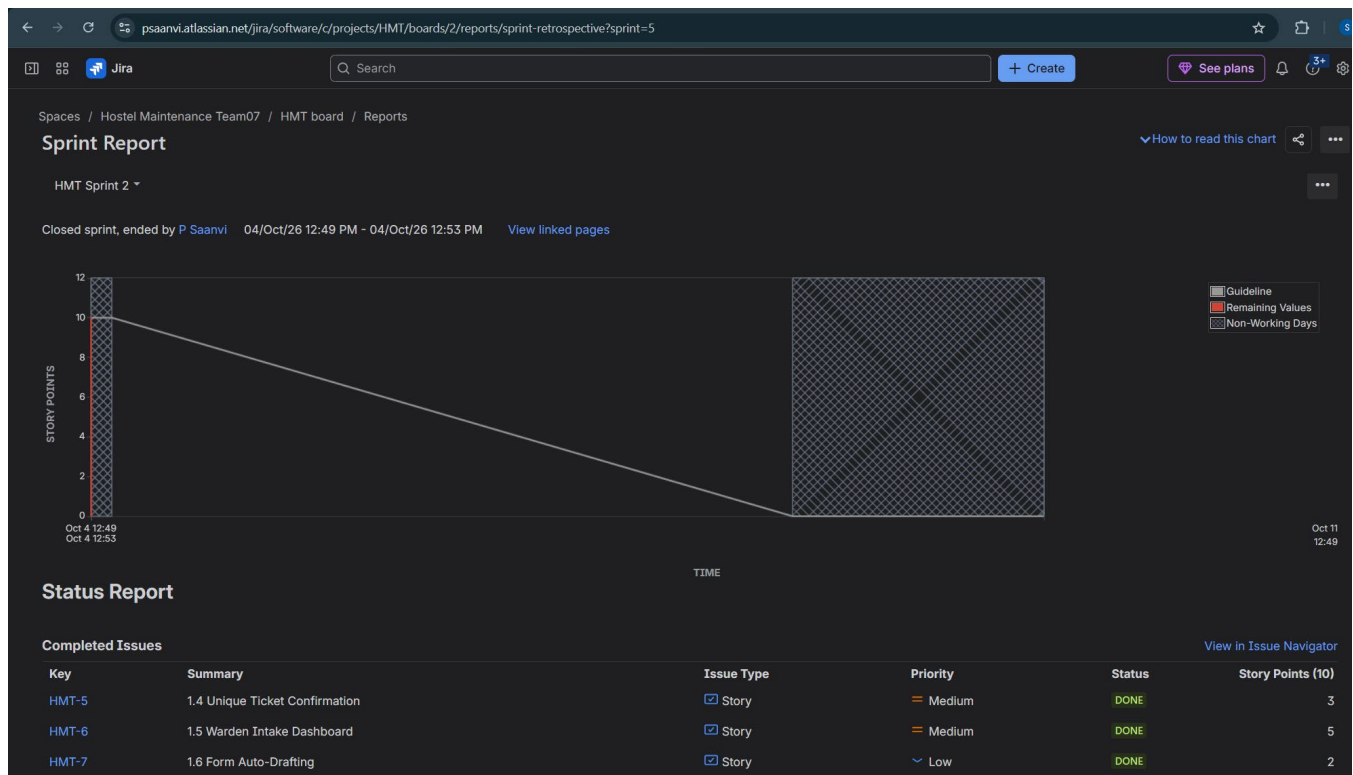


Figure 7: Sprint 2 outcome (Sprint Report, taken after completion) – all stories DONE, 10 story points.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (13 pts)

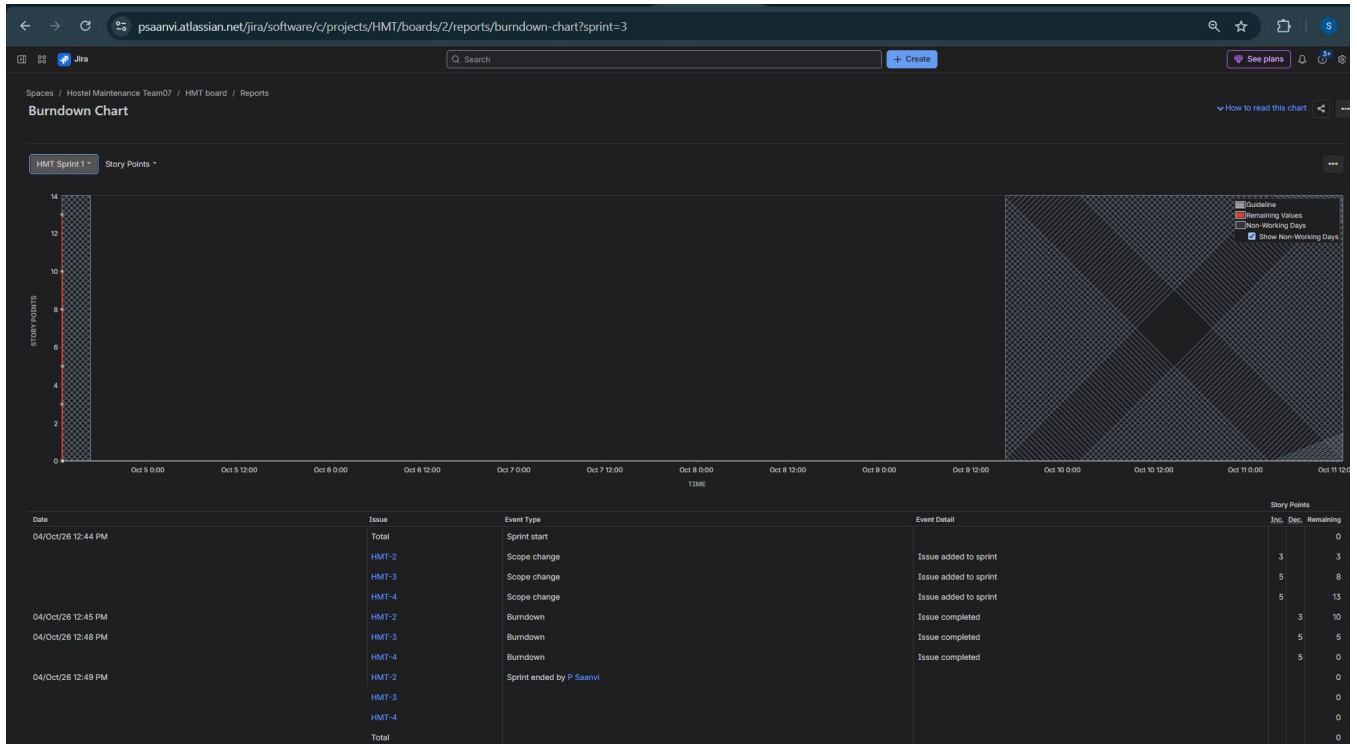


Figure 8: Sprint 1 Burndown, non-working days shown. $13 \rightarrow 0$.

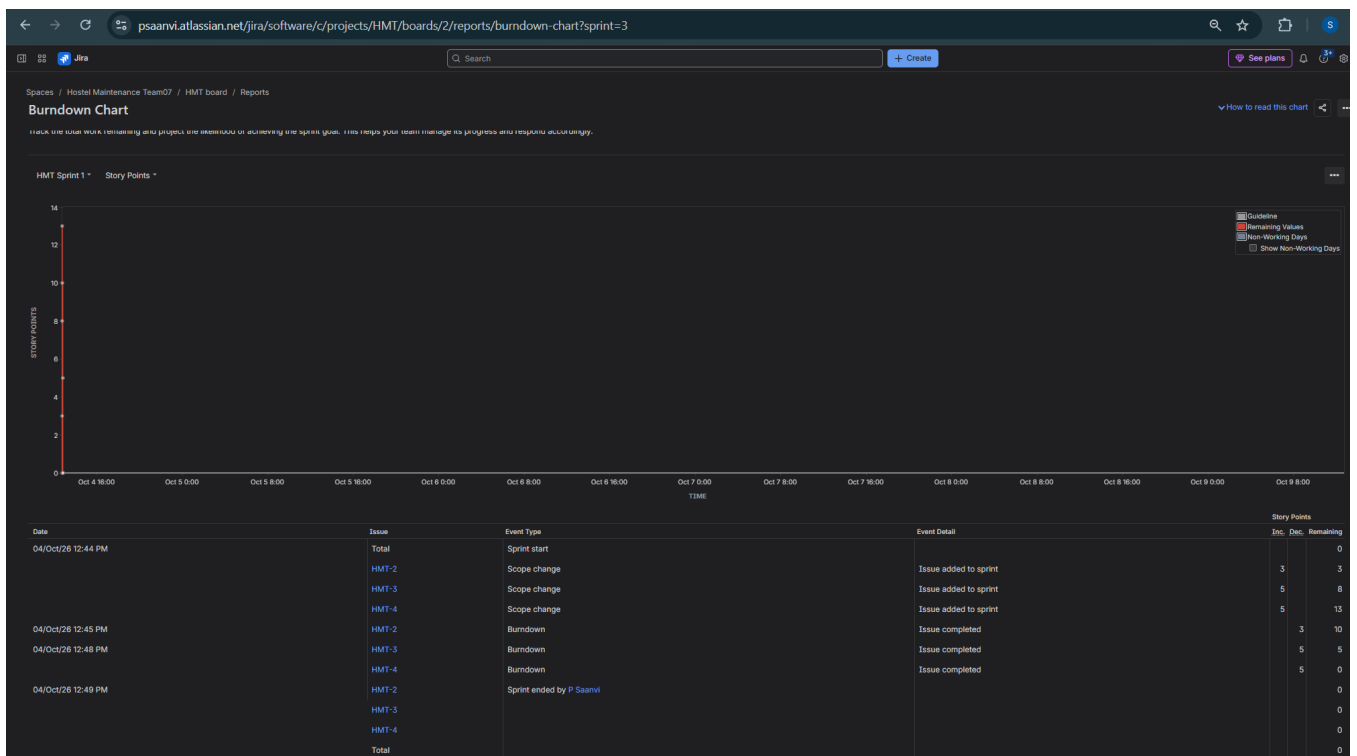


Figure 9: Sprint 1 Burndown, non-working days hidden.

Sprint 2 Burndown (10 pts)

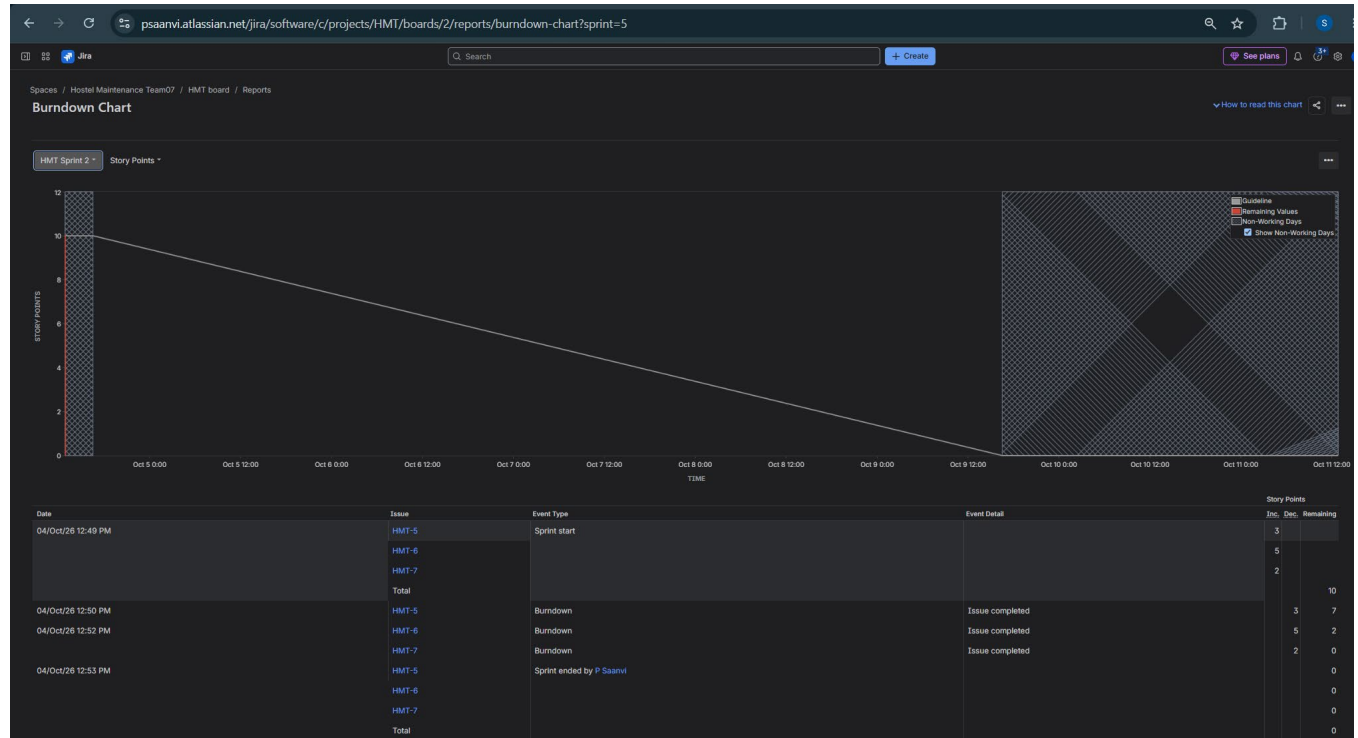


Figure 10: Sprint 2 Burndown, non-working days shown. $10 \rightarrow 0$.

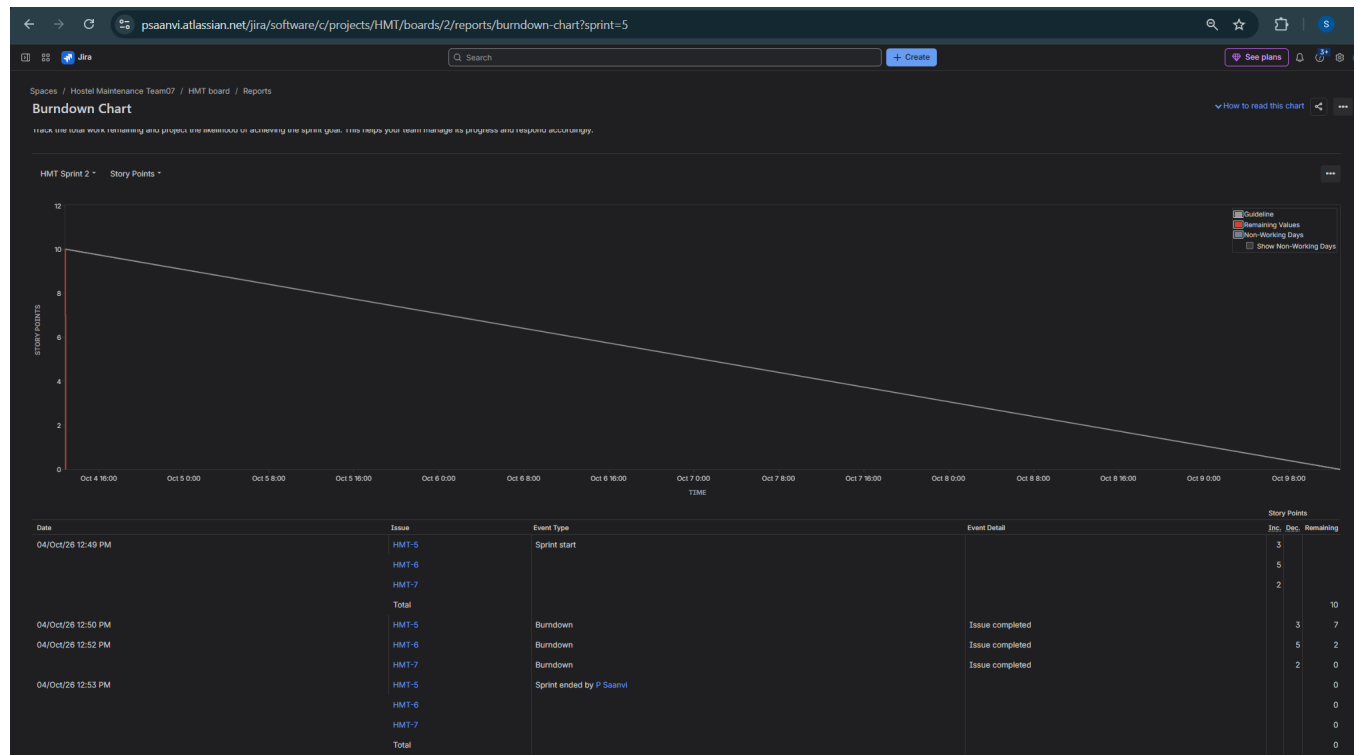


Figure 11: Sprint 2 Burndown, non-working days hidden.

Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	12:44 PM – 13	12:44 PM HMT-2 (-3) → 0 10; 12:45 PM HMT-3 (-5) → 5; 12:48 PM HMT-4 (-5) → 0	
Sprint 2	12:49 PM – 10	12:49 PM HMT-5 (-3) → 0 7; 12:50 PM HMT-6 (-5) → 2; 12:53 PM HMT-7 (-2) → 0	

8. Reflection Questions**1. Did your estimations reflect the actual effort?**

We can only partly judge this because sprints were simulated by transitioning Jira cards rather than full coding. However, relative sizing was well-calibrated: Photo Evidence Upload (5 pts) and Warden Intake Dashboard (5 pts) involved binary file handling and complex multi-column sorting queries, correctly sized higher than basic dropdown room selection (3 pts) and client auto-save (2 pts). In production, network latency on mobile photo uploads would carry the highest risk.

2. Was your backlog well-prioritized?

Yes. The three High-priority stories (1.1, 1.2, 1.3) formed the mandatory data capture contract of FR-001 and were scheduled in Sprint 1. The dashboard (1.5) was placed in Sprint 2 because an intake dashboard is useless until tickets can be logged. Auto-drafting (1.6) was kept Low as a non-blocking enhancement.

3. How did your simulated sprint align with your plan?

Both sprints executed exactly as planned: Sprint 1 committed and burned down 13 points, and Sprint 2 committed and burned down 10 points with zero carryover. Stories moved across To Do -> In Progress -> Done in order of priority.

4. What insights did the burndown chart give about your team's capacity?

The burndown showed clean step-down drops from 13 to 0 in Sprint 1 and 10 to 0 in Sprint 2. Because stories were kept to 2, 3, and 5 points, progress remained incremental. Sizing tasks at 5 points or lower prevents artificial plateaus on the chart and reflects a healthy team capacity of 10–13 points per sprint.